Monitoring, Mitigation, and Response (MMR) Plan Guidance

An MMR plan provides the framework by which a disposal scheme approval holder (i.e., operator) identifies, monitors, mitigates, and responds to induced seismicity associated with subsurface operations. The plan must demonstrate that seismic hazards are proactively managed and that timely, effective responses will be implemented when seismicity exceeds defined thresholds.

Early Seismic Hazard and Risk Identification

Directive 065 requires operators to conduct a site-specific seismic hazard assessment and, where the assessment indicates a potential for induced seismicity, to identify risks for those hazards. The assessment should inform the design of monitoring thresholds and response actions. The seismic hazard assessment should consider the following:

- A review of historical and current seismicity in the area. At a minimum, review the Alberta Geological Survey earthquake dashboard. Consider seismic activity that occurred within at least a10-kilometre radius of the site/disposal scheme.
- Geological and geomechanical conditions, with emphasis on fault presence, formation properties, and stress regimes within the targeted subsurface strata.
- Operational parameters (e.g., injection volumes, rates, pressures, and timing) that may influence seismic activity.
- The potential effects of induced seismicity on stakeholders, population centres, private property, and "essential infrastructure," as defined in the *Critical Infrastructure Defence Act*.

Consideration of Offset Activities

The seismic hazard assessment should also consider the following:

- A review of offset subsurface activities (e.g., production, enhanced recovery, disposal, carbon dioxide sequestration) occurring within the same or hydraulically connected formations.
- Consideration of whether those activities may contribute to or influence seismicity within the area of operations over the life of the proposed disposal scheme.
- Review and use of available well data or regional subsurface data (e.g., pressure and volumetric information) to inform the development of the MMR plan.
- Documentation of how the review of offset activities informed the MMR plan.

Where offset operations target different formations, joint development of the MMR and coordinated response actions with other operators may not be required unless evidence indicates a shared contribution to seismicity.

Coordination Between Operators

Where induced seismicity may be influenced by multiple operators acting within the same subsurface strata, operators are expected to take the following actions:

- Engage early with other involved operators during the development of the MMR plan.

- Establish an approach for coordinating actions should seismic activity trigger mitigation or response measures. See the Seismic Hazard Management Response section below for more on coordinated response when subsurface disposal operations include overlapping response areas.
- Include provisions outlining how information may be shared among operators (e.g., seismic data, operational changes, and response actions) to support effective risk management.

The expectation is coordinated outcomes, not prescribed operational methods. Operators retain flexibility in how coordination is achieved, provided seismic hazards are effectively addressed.

Monitoring Program

The monitoring program component of the MMR plan must address the following criteria:

- Monitoring is appropriate for the assessed level of seismic risk.
- Monitoring enables near-real-time detection and assessment of seismic events.
- Monitoring supports rapid implementation of mitigation or response actions when thresholds are exceeded.
- Seismic monitoring data is required to be submitted near-real time to the AER and the EarthScope consortium.

Seismic Hazard Management Response

The MMR plan is required to include a traffic light response framework that clearly defines the following:

- Seismic thresholds: green-, yellow-, and red-light local magnitude (ML) thresholds for each region of operation. The iso-nuisance model for the province defines the red-light thresholds, while the yellow-light thresholds are set two magnitudes lower.
- Required actions at each stop-light threshold.
- Escalation procedures for continued or increasing seismicity.

In areas of overlapping influence, response actions—particularly at yellow- and red-light thresholds—should reflect coordinated efforts where multiple operators may be contributing. In overlapping areas, magnitude thresholds should be aligned across operators.

Where multiple subsurface disposal operations contribute to reservoir stress changes, particularly increases in reservoir pressure from hydraulically connected operations, the AER expects all operators to collaborate and have a data-sharing arrangement. Shared stewardship of the subsurface reservoir is essential to managing cumulative pressure effects, reducing the likelihood and magnitude of induced seismic events, and ensuring the continued safety of the public, infrastructure, and the environment, while supporting ongoing disposal operations.

When pressure communication exists, localized seismicity may not be attributable to a single operator. Therefore, in areas of overlapping response, coordinated management is required. Operators are expected to align their MMR plans, implement compatible monitoring systems, and synchronize their traffic light protocol thresholds and response actions at an area-wide scale. When a yellow event

threshold in the traffic light protocol is exceeded, the AER expects direct communication among area operators and the execution of coordinated mitigation strategies.

Reporting of Induced Seismic Event

An operator with an MMR plan is required follow the traffic light protocol outlined in its MMR plan.

When a seismic event occurs and the magnitude of the event equals or exceeds the yellow- or red-light threshold, the operator is required to immediately report the event to the Alberta Energy Regulator through the Energy and Environmental Emergency 24-Hour Response Line at 1-800-222-6514.

Focus on Seismic Hazard Reduction

The primary objective of mitigation and response actions is to reduce the magnitude and frequency of seismic events. Attribution of causation is secondary to ensuring that effective and timely actions are taken.

Adaptive Management

Operators should review and update their MMR plans as new data, monitoring results, or operational changes become available, and should incorporate lessons learned from seismic events and coordinated response actions with other operators within shared subsurface strata.